Simulator-Based Offline RL for Sequential Movie Recommendation

Effects of Fatigue-Aware Reward Design on PPO and DQN

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Problem Motivation: Why RL for Recommendation?

The Problem

- Myopic recommenders optimize immediate clicks rather than session-level satisfaction
- Repeated same-genre exposure causes user fatigue and weakens engagement over time
- User preference naturally shifts during a session, so diversification matters
- Static or supervised models miss sequential effects across multiple recommendations

Why RL + Why Simulator

- Recommendation is a sequential decision problem that can be modeled as an MDP
- RL optimizes cumulative long-term reward instead of one-step prediction accuracy
- Learning directly from real users is expensive, slow, and risky for exploration
- A simulator enables safe offline training and evaluation before real deployment

Key takeaway We study recommendation as a long-horizon optimization problem and use offline simulation to test whether RL can learn diversity-aware, fatigue-sensitive policies.

Data Insight: Real Users Diversify, but Rewards Are Exploitable

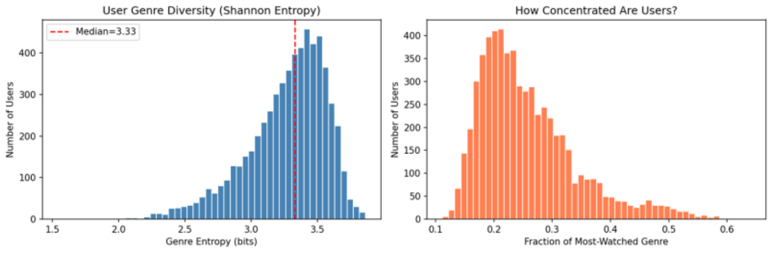
Ratings: 1M | Users: 6,040 | Movies: 3,706 | Genre: 18

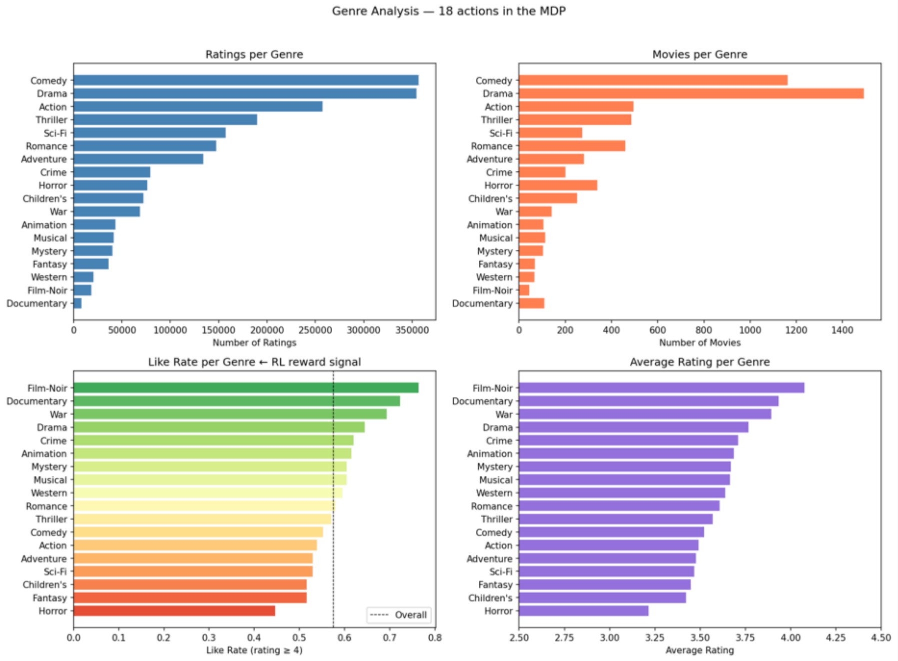
Users Naturally Diversify

- Genre entropy: 3.27 / 4.17 bits (high diversity)
- Median dominant genre: only 24%
- Real users don't spam one genre

Genre Like-Rates Vary Widely

- Horror: 44.6% → Film-Noir: 76.3%
- 30pt spread across genres
- An exploit-prone reward landscape





The data suggests a mismatch: real users diversify, but the reward landscape still encourages genre

MDP Setup for Genre-Level Recommendation

State	68-dim: SVD user embedding (50) + recent genre counts (18)
Action	18 genre indices; env selects best movie within genre
Reward	*No-fatigue:* *Bernoulli(P(like))* *Fatigue-aware:* *Bernoulli(clip(P(like) – fatigue_penalty, 0.1, 0.9))*
Episode	T = 20 recommendation steps per user session
Transition	Genre counts update via sliding window (w=10)

Hypothesis: Does the Simulator Capture Fatigue?

If the learned simulator captures fatigue, predicted like probability should decline as the same genre repeats in recent history.

Probe Result: Under our genre-level probe, we do not observe a clear, recoverable fatigue signal

- Repeating the same genre does not meaningfully reduce predicted like probability
- Mean P(like) is nearly flat from $k = 0$ to 4 ($\Delta = 0.009$)
- This may reflect limited identifiable fatigue signal in the historical data, rather than a pure simulator failure
- Without an explicit repetition cost, high-like-rate genres remain profitable and can be over-selected by RL policies

No Artificial Fatigue: Real-Data Evaluation & Cross-Simulator Robustness

Does RL still outperform Greedy without artificial signals?

Simulators: GRU4Rec + XGBoost

Cross-project model transfer: independently trained GRU4Rec with SVD-pretrained embeddings

GRU4Rec (Phase 1)	GRU4RecSVD (Phase 2)	XGBoost (Reference)
AUC = 0.703	**AUC = 0.781**	**AUC = 0.800**
Random init (50d) T=20 sequence Sigmoid output	SVD-64 pretrained T=100 sequence Logit + external σ	Static features No sequence Highest pointwise AUC

GRU4RecSVD improves pointwise accuracy while preserving sequential structure for RL experiments.

Results Without Artificial Fatigue

No penalty applied — reward = Bernoulli(P(like)) from simulator only

Policy	Original GRU4Rec (AUC 0.703)	Enhanced GRU4RecSVD (AUC 0.781)	Rank Preserved?
Random	10.05 ± 2.20	9.49 ± 1.78	✓
Greedy-CTR	11.33 ± 2.43	10.88 ± 2.62	✓
DQN	12.27 ± 3.12	12.25 ± 2.20	✓
PPO	13.38 ± 2.79	12.63 ± 1.93	✓

Takeaway

- **Ordering is preserved across simulators: PPO > DQN > Greedy > Random**
- **The PPO–DQN gap shrinks from 1.11 to 0.38 with the stronger simulator**

Possible explanation: better simulator quality reduces noise, which may shrink PPO's stability advantage over DQN.

Why RL > Greedy Without Fatigue?

Fatigue Probe (Enhanced GRU4RecSVD)

k (repeats)	Mean P(like)
0	0.451
1	0.449
2	0.451
3	0.463
4	0.440

Result: No obvious fatigue signal ($\Delta = 0.023$, Std ≈ 0.24)

Even without explicit fatigue, RL can outperform Greedy because the simulator remains sequence-conditional: RL can optimize action ordering over time, while Greedy only maximizes immediate reward.

Three-Part Explanation

Greedy is Myopic

Picks highest P(like) each step. Ignores effect on GRU hidden state.

GRU is Sequence-Conditional

Same movie gets different P(like) after different sequences.

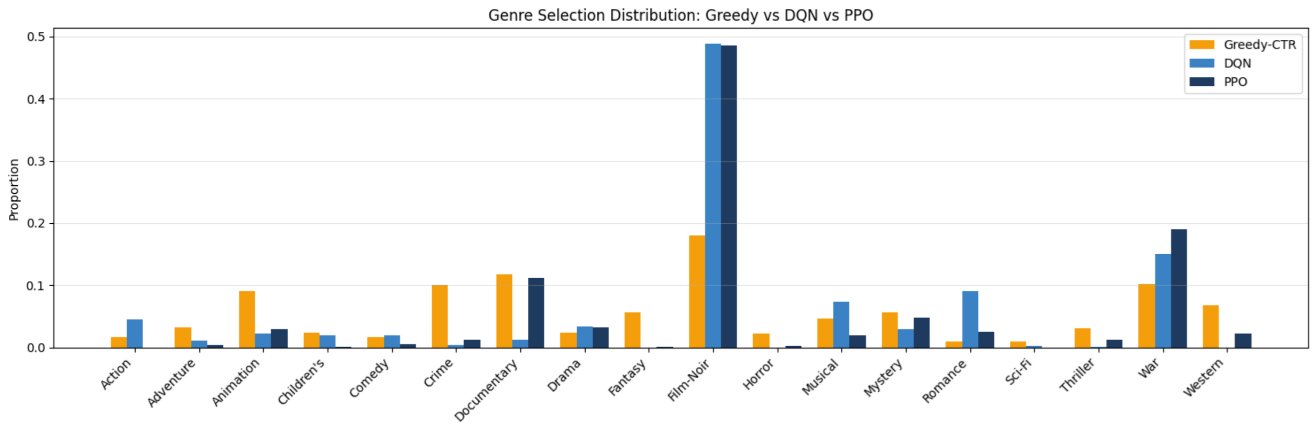
RL Does Sequential Planning

PPO learns genre orderings that maximise cumulative reward.

Honest Caveat: Cannot confirm whether GRU sequence dynamics reflect real user patterns or model-induced noise. AUC = 0.78 still means substantial prediction error. Definitive validation requires online A/B testing or session-based datasets.

Genre Distribution: RL Exploits Simulator Bias

Without fatigue penalty, RL agents exploit simulator bias instead of learning genuine diversification



Greedy Is Less Exploitative

Greedy optimizes only immediate reward, Greedy spreads recommendations more broadly and does not learn the same single-genre exploitation pattern as RL.

Film-Noir Collapse

PPO and DQN each allocate 49% of selections to Film-Noir. Despite being a small genre (44 movies), it receives persistently high predicted reward in the simulator with little decay under repetition.

Without fatigue penalty, RL exploits simulator bias. Explicit fatigue modeling helps reduce this exploitation and better aligns policy behavior with observed user diversification.

Solution: Explicit Fatigue Penalty

Window Penalty

Penalizes recent same-genre repetition within a sliding window of 10 steps

```
min(0.05 × same_count, 0.25)
```

Concentration Penalty

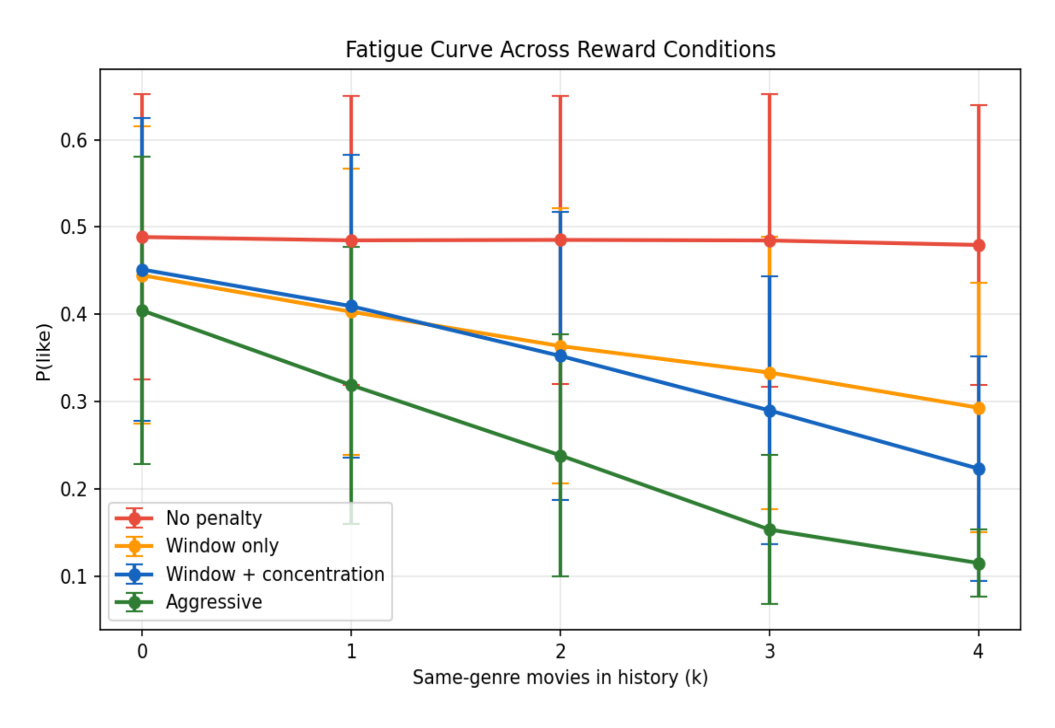
Penalizes session-level genre dominance above 25% share of total picks

```
0.3 × max(genre_frac - 0.25, 0)
```

Why these penalties?

- Data shows real users diversify (entropy 3.27/4.17, dominant genre only 24%)
- GRU4Rec doesn't capture this → explicit penalty bridges the gap
- Window penalty handles short-term repetition, concentration handles session-level exploitation

Fatigue Signal Across Reward Conditions



No penalty

Δ = 0.009

Window only

Δ = 0.152

Window + conc.

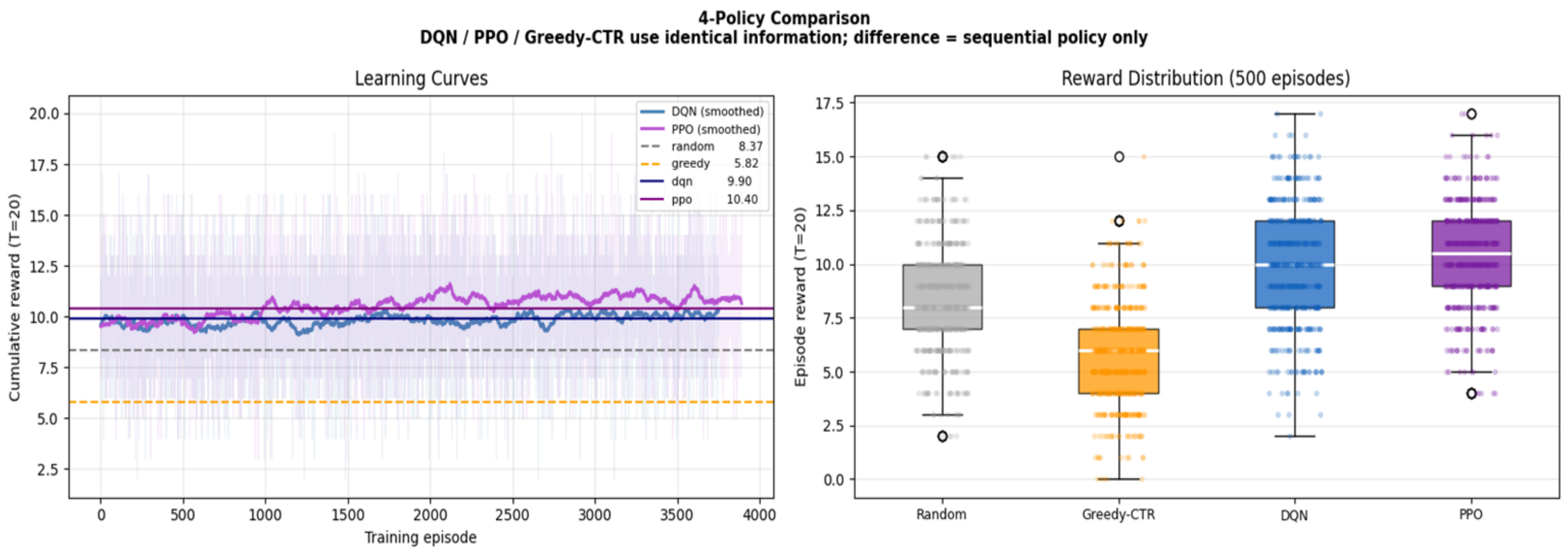
Δ = 0.228

Aggressive

Δ = 0.289

Stronger penalty → steeper fatigue curve → more signal for RL agents

Results: Policy Comparison (Window + Concentration)



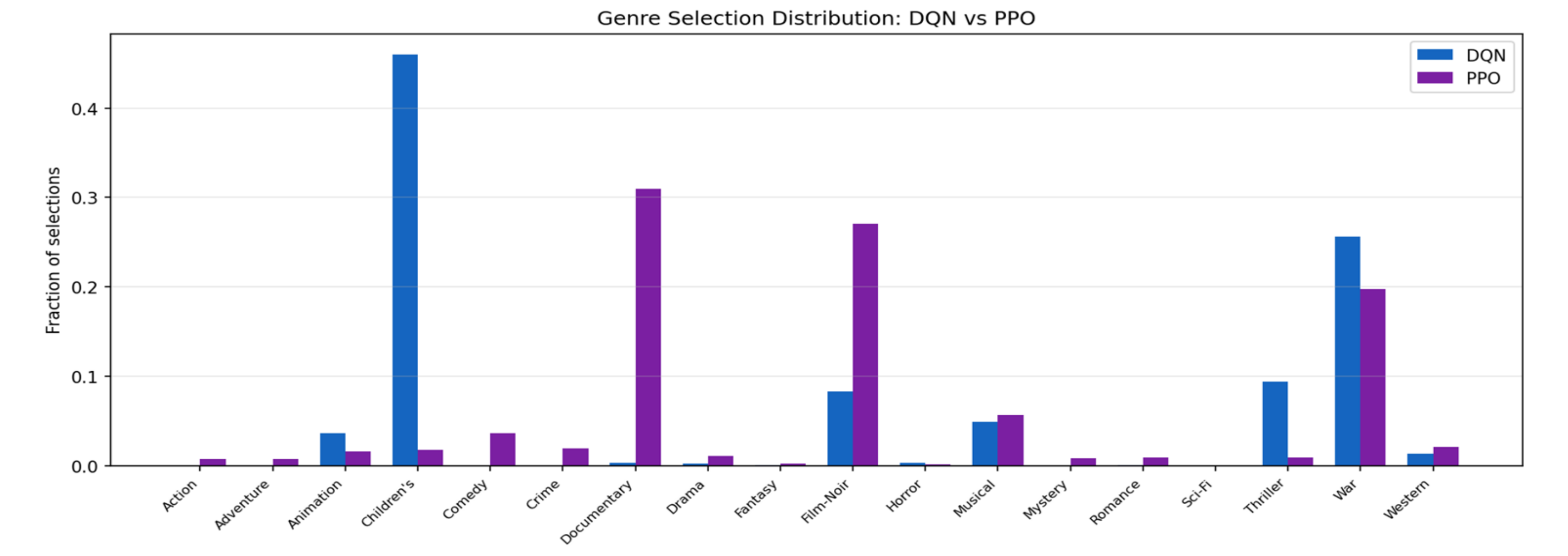
PPO vs Greedy-CTR: +4.31

$p < 0.001$, significant

DQN vs Greedy-CTR: +1.55

$p < 0.001$, significant

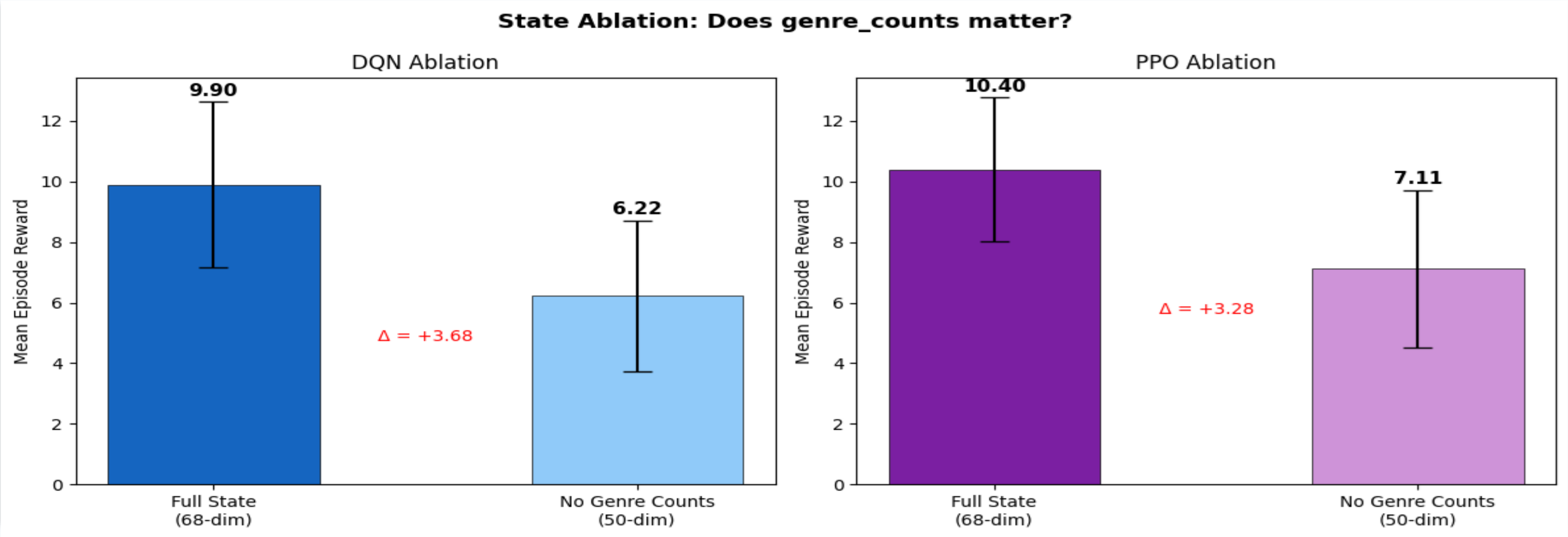
Genre Selection: DQN vs PPO



DQN: 46% Children's
Deterministic policy → concentrates on one genre

PPO: 31% Doc, 27% Film-Noir, 20% War
Stochastic policy → reduces single-genre collapse

Ablation: Does genre_counts Matter?



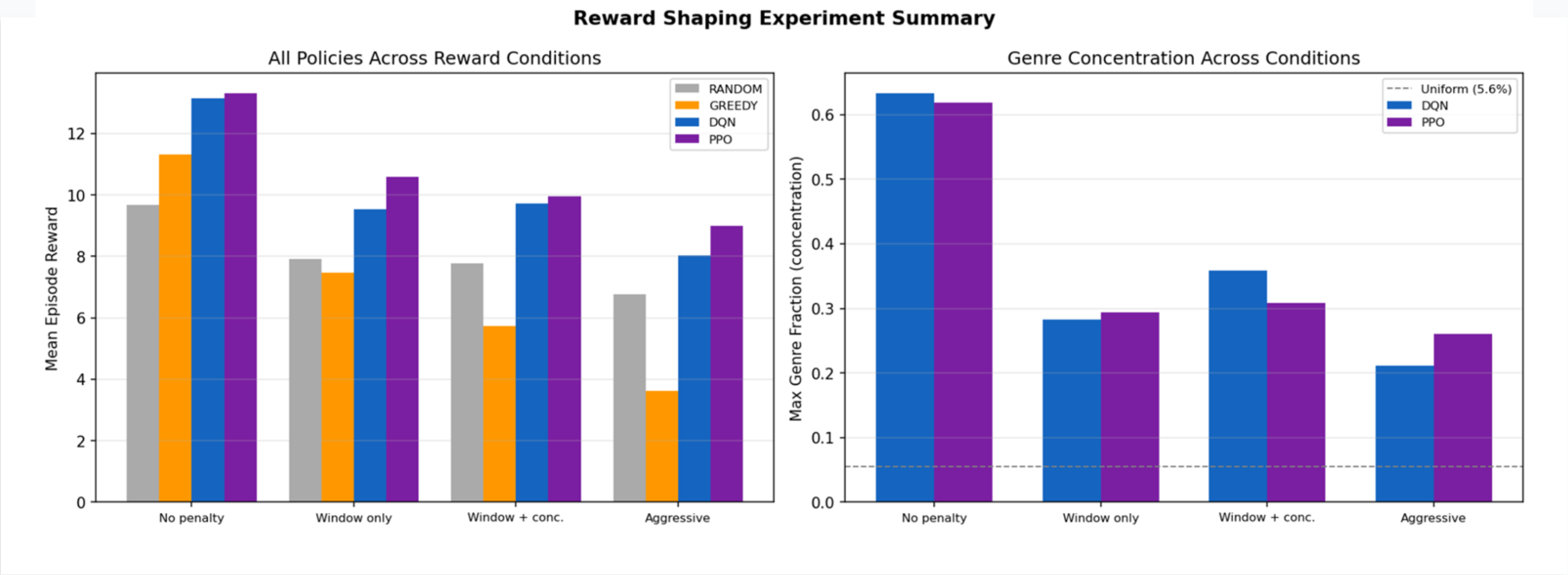
DQN: Δ = +2.05

7.22 (full) → 5.17 (no genre counts)

PPO: Δ = +4.43

9.98 (full) → 5.55 (no genre counts)

Reward Shaping: 4 Conditions Compared



PPO vs Greedy advantage grows with penalty strength: +1.99 → +3.11 → +4.23 → +5.37

PPO adapts to harder constraints; Greedy-CTR collapses (11.31 → 3.63)

Limitations & Future Work

Limitations

Fatigue signal is weak and probe-dependent

Under our genre-level probe, we do not observe a clear recoverable fatigue signal in the learned simulator.
This may reflect limited identifiable fatigue signal in MovieLens rather than a pure simulator failure.

Simulator bias can still drive unrealistic policies

Without explicit penalties, RL can exploit over-rewarded genres instead of learning realistic diversification.

Offline results may not transfer fully online

Policy trained on simulator may underperform on real users — offline evaluation cannot fully validate live performance

Cold-start and item generalization remain limited

Our simulator is movie-ID based, so it is not a true cold-start model for unseen movies.

Future Work

Switch to KuaiRec / Tenrec

Short-video datasets with skip/watch-ratio signal — genuine fatigue structure matching TikTok/Kuaishou production scenario

Improve item representation for cold-start

Replace pure movie-ID inputs with hybrid item features, such as genre multi-hot, metadata, or pretrained content embeddings.

Refine fatigue probing for multi-genre movies

Instead of sampling one target genre, future probes could use multi-genre overlap statistics to better capture repeated-exposure effects.

Online or human-in-the-loop validation

Validate whether fatigue-aware reward shaping improves real sequential experience beyond offline simulator gains.

Industry Relevance: Our findings on simulator bias, data limitations, and offline-to-online mismatch are directionally aligned with challenges discussed by large-scale recommendation platforms such as Netflix, YouTube, and ByteDance.

Key Findings

1. No strong genre-fatigue signal is recoverable from the learned simulator, suggesting limited fatigue evidence in the data.
2. Without fatigue penalty, RL outperforms Greedy partly by exploiting simulator bias.
3. Explicit fatigue penalties reduce single-genre exploitation and change policy behavior substantially.
4. Under fatigue-aware rewards, PPO learns a less concentrated policy and outperforms DQN and Greedy.
5. Reward design matters at least as much as algorithm choice in this setting.

Thank You

Questions & Discussion

https://github.com/Bryan-bits/Fatigue-Aware-Recsys